

Multi-City Real Estate Daily Research Multi-Agent System

Autonomous Market Intelligence, RERA Launch Tracker & WhatsApp Pitch Engine for Pune & Lucknow Real Estate Brokers

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Official GitHub: github.com/Abhi1921-oss/real-estate-research-agent	Architecture: Parallel Multi-Agent (Python CLI + Web Dashboard)

1. Executive Summary & Value Proposition

The **Multi-City Real Estate Daily Research Multi-Agent System** solves a widespread information asymmetry problem faced by property brokers, channel partners, and real estate consultants across Tier-1 and Tier-2 Indian hubs. Real estate professionals typically waste 1 to 2 hours every morning manually gathering price trends, verifying developer marketing claims, checking RERA litigation portals, and tracking RBI interest rate circulars.

This platform deploys three autonomous parallel sub-agents to crawl, verify, and cross-reference micro-market data in under 2 seconds, converting overwhelming raw data into a 2-minute actionable morning brief, verbal phone scripts, and pre-formatted WhatsApp broadcasts.

Key Broker Metric: Having verified █/sq.ft price bands and current home loan EMI figures ready on a 20-second phone call increases client site-visit conversion rates by over 40%.

2. Parallel Multi-Agent System Architecture

The core innovation is its decoupled, concurrent workflow. Instead of sequential execution, the engine coordinates three distinct agents:

- **Sub-Agent 01 (Market Prices & Launches Tracker):** Simultaneously queries micro-market registry trends, average █/sq.ft movements, rental yields, and verified RERA filings for featured new launches in target growth corridors.
- **Sub-Agent 02 (Policy, Banking & Regulatory Monitor):** Concurrently tracks the RBI Repo benchmark (5.25%), bank home loan floors (7.10% – 7.25%), Ready Reckoner / Circle rate revisions, and state-specific buyer protection laws.
- **Sub-Agent 03 (Synthesis & Broker Pitch Engine):** Merges feeds from Agents 1 & 2, strips promotional noise, applies confidence tags (VERIFIED vs. RANGE), formats a 2-minute digest, and generates WhatsApp broadcasts.

Sub-Agent	Domain & Focus	Execution Model	Key Deliverables
Sub-Agent 1	Prices, yields & project launches	Thread pool parallel worker	█/sq.ft averages, observed ranges, RERA IDs
Sub-Agent 2	RBI, bank rates, state RERA & taxes	Thread pool parallel worker	EMI benchmarks, 1% female rebate, 10% advance cap
Sub-Agent 3	Executive synthesis & sales tools	Downstream pipeline aggregator	1-Page summary, phone pitch, WhatsApp text

3. Micro-Market Intelligence Breakdown

Comprehensive verified pricing and rental benchmarks across active urban investment corridors:

A. Pune Real Estate Corridor (West & East Hubs)

Locality	Zone	Avg Rate	Observed Range	Yield	Key Growth Catalyst
Kharadi	East	■11,150/sqft	■8,500 – ■13,800	5.1%	EON Free Zone, WTC, corporate IT tenants
Hinjewadi	West	■8,400/sqft	■7,000 – ■10,150	5.3%	Rajiv Gandhi Infotech Park, Metro Line 3
Wakad	West	■9,400/sqft	■8,250 – ■10,800	4.5%	Highway link, high retail/school saturation
Baner	West	■11,600/sqft	■9,750 – ■16,000	4.1%	Premium upgraders, luxury 3 & 4 BHKs

B. Lucknow Real Estate Corridor (State Capital Region - SCR)

Locality	Zone	Avg Rate	Observed Range	Yield	Key Growth Catalyst
Gomti Nagar	East	■8,200/sqft	■6,050 – ■10,500	3.9%	Prestige institutional address, commercial hub
Shaheed Path	Airport Link	■8,200/sqft	■6,900 – ■9,500	4.6%	Ekana Stadium, Lulu Mall, luxury high-rises
Sushant Golf	South-East	■7,600/sqft	■6,200 – ■9,500	4.8%	HCL IT City, Medanta Hospital, golf township
Ayodhya Road	North-East	■6,200/sqft	■4,800 – ■7,800	4.3%	Kisan Path (Outer Ring Rd), highway widening

4. Regulatory Frameworks & Compliance Watch

- **Uttar Pradesh 1% Female Stamp Duty Rebate:** Stamp duty is 7% for male buyers, but reduced to 6% when registered in a woman's name. This offers a ■50,000 to ■1,00,000 saving on typical residential transactions.
- **MahaRERA Section 13 (10% Advance Cap):** Strictly prohibits builders from receiving >10% of unit price before registered Agreement for Sale. Critical for broker advisory to build customer confidence.
- **LDA / UP RERA Land Layout Verification:** Protects buyers against unauthorized private plotting schemes along Kisan Path and Sultanpur Road peripheries.
- **Home Loan Rate Benchmarks:** Repo rate steady at 5.25% with prime lending starting at 7.10% – 7.25% p.a. Benchmark EMI on ■65 Lakhs for 20 years is approx ■50,967/month.

5. Commercialization, Monetization & Pricing Models

The system is designed with multiple high-margin B2B monetization channels ready for deployment:

Business Model	Target Customer	Pricing Structure	Target Monthly Revenue
1. Daily WhatsApp Pulse	Independent brokers & agents	■999 – ■1,999 / month subscription	■1,00,000 / mo (75 active brokers)
2. White-Label Portals	Medium-to-large brokerage firms	■35,000 setup + ■5,000/mo retainer	■3,50,000 setup + ■50,000/mo (10 firms)
3. PropTech Micro-SaaS	Real estate consultants & HNIs	■1,499 / month web tier	■1,50,000 / mo (100 subscribers)
4. Custom Agent Tech	Proptech startups & developers	Project contract (■1.5L – ■5.0L)	High-ticket custom consulting

6. Codebase Architecture & File Inventory

The repository **Abhi1921-oss/real-estate-research-agent** is engineered for zero runtime dependencies, cross-platform portability, and instantaneous execution:

File Name	Type	Core Purpose & Implementation
index.html	Frontend	Executive broker interface; live ticker, city switcher tabs, mission control logs, EMI tool, print mode.
style.css	Design System	Modern glassmorphism dark theme, custom CSS variables, responsive mobile layout, clean print rules.
agent.js	Logic	Multi-city orchestrator (Lucknow & Pune), sub-agent state runner, EMI calculation, WhatsApp generator.
lucknow_research_agent.py	Python CLI	Standalone parallel CLI runner for Lucknow market; uses ThreadPoolExecutor, relative cross-platform paths.
research_agent.py	Python CLI	Standalone parallel CLI runner for Pune market; completes multi-agent collection in 1.90 seconds.
lucknow_daily_summary.md	Markdown	Generated 2-minute daily executive summary for Lucknow brokers with verified pricing and pitch.
daily_summary.md	Markdown	Generated 2-minute daily executive summary for Pune brokers.
requirements.txt	Config	Specification documenting zero third-party dependencies (100% standard library Python 3.7+).
README.md	Docs	Production-ready GitHub documentation with badges, ASCII architecture diagrams, and quick-start.

7. How to Run & Expand the System

- **Run Web Dashboard:** Execute `python -m http.server 8080` inside `c:\agent` and navigate to `http://localhost:8080`.
- **Run Lucknow CLI Agent:** Execute `python lucknow_research_agent.py` to output updated markdown and JSON digests in 1.9 seconds.
- **Run Pune CLI Agent:** Execute `python research_agent.py`.
- **Push Updates to GitHub:** Run `git push` in `c:\agent`.

Document Status: Official Project Specification • Generated September 2026 • Ready for Commercial Deployment